

Venkata Sai Kandipati

SENIOR JAVA DEVELOPER | SPRING BOOT & CLOUD

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AWS CERTIFIED AI PRACTITIONER

PROFESSIONAL SUMMARY

- 7+ years of experience designing, developing and supporting highly available, scalable and reliable backend services and microservices in Java and Spring Boot.
- Strong experience with distributed systems and event processing — independently deployable microservices communicating over REST, messaging and event-driven flows, designed for high availability and graceful degradation.
- Good exposure to both monolithic and microservices architectures, with hands-on containerisation and orchestration using Docker and Kubernetes, including deployments, services and load balancing.
- Proficient with the Spring ecosystem — Spring Boot, Spring MVC, Spring Data JPA, Spring Security and Spring AOP — with Hibernate, JPA and JDBC for persistence.
- Hands-on experience with relational and NoSQL databases — PostgreSQL, Oracle, MySQL, SQL Server and MongoDB — including schema design, complex SQL, indexing and query performance tuning.
- Hands-on AWS experience with Lambda, Step Functions, SQS, SNS, CloudWatch, Secrets Manager, EC2, S3 and RDS, supporting cloud-native and event-driven applications.
- Hands-on experience with Apache Kafka, RabbitMQ, Microsoft Azure, Git, Maven, Jenkins, GitHub Actions and Azure DevOps pipelines for continuous delivery, build and deployment.
- Strong experience with automated testing, CI/CD, production troubleshooting, code reviews, API security, database design and performance optimisation.
- Good exposure to TDD and automated unit, integration and regression testing with JUnit and Mockito as part of every change.
- Experience across banking, payments and enterprise platforms, applying cloud design principles and strict security, authorisation and data privacy requirements with JWT and role-based access control.
- Practical experience using AI-assisted development tools including Claude, Cursor and GitHub Copilot for development, debugging, refactoring, testing and technical documentation, backed by AWS Certified AI Practitioner certification.
- Full-stack exposure with JavaScript, React and Angular alongside enterprise backend development.
- Good understanding of Agile Scrum delivery, SDLC and software design principles, with strong analytical, problem-solving and requirement-analysis skills.

CORE COMPETENCIES

End-to-End Solution Ownership | Automated Testing & Quality Assurance | Stakeholder Engagement in Consulting Environments | Code Review & Engineering Standards | Backend Service Design & Development | Team-Based Delivery & Sprint Ceremonies | Front-End Contribution Alongside Backend Delivery | Independent Delivery & Professional Accountability | Service Integration & Interface Contracts | Legacy System Modernisation & Technical Debt Reduction | Relational Database Design & Data Modelling | Technical Documentation & Handover | Feature Delivery Within Established Architectures | Application Security & Authorisation | Cloud Adoption & Migration Support | Debugging & Root-Cause Problem Solving | Efficient SQL & Query Performance Tuning

TECHNICAL SKILLS

- **Core Java:** Java, Collections, Streams, Lambda expressions, Generics, exception handling, object-oriented programming (OOP), SOLID principles, design patterns
- **Backend & APIs:** Spring Boot, Spring Framework, Spring MVC, Spring Data JPA, Spring AOP, REST APIs, RESTful API design, dependency injection / IoC, SOAP web services, Swagger / OpenAPI, RESTful web services, Apache Kafka, Microservices, Event-Driven Architecture, API Integration, High Availability, Distributed Systems, Event Processing, RabbitMQ
- **Persistence & Web Tier:** Hibernate, JPA, JDBC, HQL / JPQL, Java EE / J2EE, Servlets, JSP, JSON, XML, XSD, Jackson
- **Front-End:** JavaScript, React, Angular, HTML5, CSS3
- **Cloud & AWS:** AWS Lambda, Step Functions, SQS, SNS, CloudWatch, Secrets Manager, EC2, S3, RDS, Aurora, Microsoft Azure, cloud design principles, load balancing, auto-scaling
- **FinTech & Integrations:** Payment Processing, Transaction Routing, DebiCheck, Reconciliation, Payment Gateways, AVS, Third-Party API Integration
- **Databases:** MySQL, Oracle, SQL Server, DB2, MongoDB, Schema Design, Data Modelling, Indexing, Query Optimisation, PostgreSQL, complex SQL, NoSQL, joins and aggregation

- **Testing:** JUnit, Mockito, Unit Testing, Regression Testing, TDD, Test-Driven Development (TDD), Integration Testing, Postman, Bruno, API Testing
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, GitHub Actions, Azure DevOps Pipelines, Maven, Nexus, Git, GitHub, Bitbucket
- **Security & Reliability:** Spring Security, Role-Based Access Control (RBAC), AWS Secrets Manager, JWT, Resilience4j, Retry, Circuit Breaker, Rate Limiter
- **Production & Observability:** Log Analysis, SLF4J / Logback, Linux (RHEL / Ubuntu), Windows, JBoss, Bash / shell scripting, Production Support, AWS CloudWatch, Root-Cause Analysis, Apache Tomcat, IBM WebSphere
- **AI & AI-Assisted Development:** Cursor, GitHub Copilot, Spring AI, LLM API Integration, Prompt Engineering, Amazon Bedrock, Claude, foundation models, retrieval-augmented generation (RAG), model evaluation, responsible AI
- **Practices & Tools:** Agile / Scrum, SDLC, Jira, Confluence, IntelliJ IDEA, Eclipse, DBeaver

PROFESSIONAL EXPERIENCE

Senior Java Developer — TechValley (Pty) Ltd

Jun 2024 – Present

Johannesburg, South Africa | Project: Transaction Workflow Orchestrator

A payment systems platform for enterprise financial clients covering payment processing, transaction routing, receipting, reconciliation and cash-flow management, engineered for reliability and scale.

Key Responsibilities:

- **Designed, developed, tested and maintained backend services** in Java and Spring Boot across the platform's payment, routing and reconciliation domains, delivering features within the established system architecture.
- **Built the platform as independently deployable microservices** communicating over REST and AWS SQS/SNS messaging, so one payment flow could be released without redeploying the rest.
- **Built and integrated services with the relational database** using Spring Data JPA and Hibernate over PostgreSQL — entity mapping, schema design and transaction management — keeping financial records consistent, auditable and recoverable.
- **Wrote and tuned efficient SQL** profiling slow paths and refining queries, indexes and transaction handling with the DBA team so response times held as transaction volumes grew.
- **Integrated third-party payment services over secure REST APIs** payment gateways, DebiCheck and AVS verification, automating end-to-end payment validation for enterprise clients.
- **Wrote automated unit and integration tests** in JUnit and Mockito across the transaction and reconciliation services, with regression checks before each release so defects surfaced well before client environments.
- **Took ownership of technical issues through to completion** troubleshooting escalated production incidents through application log and metric analysis in CloudWatch and command-line investigation on Linux hosts, then delivering durable fixes rather than repeat patches.
- **Applied resilience patterns for high availability** Retry, Circuit Breaker and Rate Limiter at the service layer, so a failing downstream provider degrades the workflow instead of stopping it.
- **Protected financial data end to end** enforcing JWT and role-based access control, and moving credentials out of source control into AWS Secrets Manager.
- **Supported cloud adoption** building event-driven orchestration on AWS Step Functions and Lambda, and containerising services with Docker for Kubernetes, built and released through Azure DevOps pipelines.
- **Delivered in two-week sprints** estimating work in planning, reporting progress through stand-ups and refinement, and carrying retrospective actions into the following sprint.
- **Collaborated with developers, analysts, testers and stakeholders** on REST API design and development, publishing contracts in Swagger/OpenAPI, reviewing code and keeping interface specifications current for the teams integrating.
- **Clarified requirements with business analysts and QA** translating them into technical specifications, and handled build, deployment and production support activities after release.

Technical Environment: Java, Spring Boot, Spring Data JPA, Hibernate, REST APIs, PostgreSQL, AWS Lambda, Step Functions, Secrets Manager, CloudWatch, Docker, Kubernetes, Azure DevOps Pipelines, Swagger/OpenAPI, JUnit, Mockito, Git, Linux.

Hyderabad, India | Project: Risk Manager

An enterprise risk-analysis platform used by global development teams to surface application vulnerabilities through static and dynamic code analysis, automating risk assessment, remediation tracking and compliance reporting.

Key Responsibilities:

- **Implemented new features within an established architecture** extending the platform's Java and Spring Boot layer over MySQL without disrupting the systems already consuming it.
- **Designed the schemas and the queries behind them** behind risk scoring and the reports generated from it, keeping output accurate and responsive under load.
- **Secured sensitive vulnerability and compliance data** with Spring Security and JWT, restricting every user to the records their role permits.
- **Specified and validated the REST endpoints** checking them continuously with Postman so integrations stayed consistent across services.
- **Debugged and resolved production defects** affecting risk assessment accuracy and compliance reporting logic, tracing each to its source and correcting the logic behind the figures.
- **Built reusable business services and shared components** using object-oriented programming and SOLID design principles, cutting duplication across the codebase and shortening delivery of related features.
- **Built internal tools and utilities to streamline development** shared helpers, test-data generators and build scripts that removed repeated manual work.
- **Drove quality through review and automated coverage** JUnit and Mockito suites, peer code review, and CI/CD maintained on Git, Maven and Jenkins.
- **Carried out unit testing, bug fixing and production support** across releases, analysing reported issues and correcting defects in the business logic behind them.

Technical Environment: Java, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, MySQL, REST APIs, JUnit, Mockito, Postman, Maven, Jenkins (CI/CD), Git, Linux.

Junior Java Developer — Genpact

Mar 2019 – Mar 2022

Hyderabad, India | Project: OpsHub

A centralised operations platform integrating GE Digital's Historian and iFIX products to collect, process and visualise industrial data, delivering real-time operational insight to manufacturing teams.

Key Responsibilities:

- **Assisted in developing backend features** in Java, Spring Boot, JPA and Hibernate for industrial data processing and reporting, deployed to Apache Tomcat on Linux servers.
- **Wrote and tested SQL queries against MySQL** to power the retrieval and reporting layer behind the monitoring dashboards manufacturing teams relied on.
- **Supported integration of GE Historian and iFIX systems** for real-time data collection, learning how the underlying measurements were produced alongside senior engineers.
- **Worked to the project's engineering standards** following its coding conventions, review process and release checks on every change.
- **Carried out API testing with Postman on every REST endpoint** ahead of each release, and maintained a documented test-case record that reduced late-stage defects.
- **Investigated and reported defects found during release validation** reproducing each issue, narrowing it to the component responsible and handing it over with enough detail for a fix.
- **Took part in deployments, environment validation and defect resolution** on Linux servers with senior engineers and QA, contributing to day-to-day platform stability.

Technical Environment: Java, Spring Boot, JPA, Hibernate, MySQL, REST APIs, Apache Tomcat, Linux, Maven, Git, Postman.

EDUCATION

Acharya Nagarjuna University – India
BCom (Computer Applications)